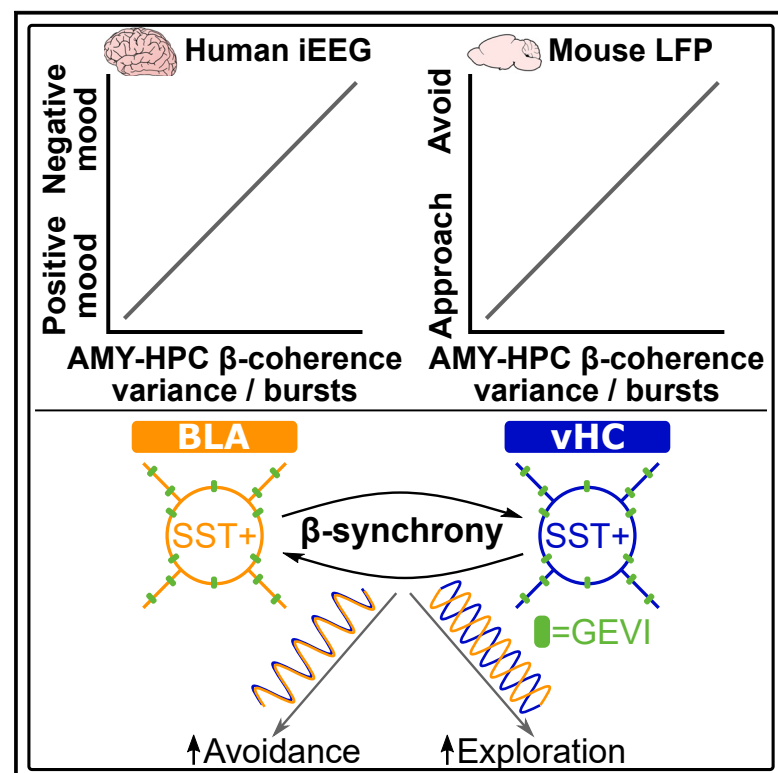


Amygdala-hippocampus somatostatin interneuron beta-synchrony underlies a cross-species biomarker of emotional state

Graphical abstract



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In brief

Examining a synchrony-based biomarker for emotional state previously identified in humans, Jackson et al. validate the same biomarker in mice, finding that it depends on amygdala and hippocampus somatostatin neurons, which are linked by direct connections. They show that this biomarker is a causal driver (not just readout) of behavior.

Highlights

- Amygdala-hippocampus (HPC) β -synchrony tracks emotional state in humans and mice
- This biomarker reflects synchronization between SST+ neurons in the BLA and HPC
- Manipulating BLA-HPC SST+ neuron β -synchrony bidirectionally alters anxiety behavior
- BLA-SST+ neurons directly inhibit ventral HPC SST+ interneurons



Article

Amygdala-hippocampus somatostatin interneuron beta-synchrony underlies a cross-species biomarker of emotional state

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SUMMARY

Emotional responses arise from limbic circuits including the hippocampus and amygdala. In the human brain, beta-frequency communication between these structures correlates with self-reported mood and anxiety. However, both the mechanism and significance of this biomarker as a readout vs. driver of emotional state remain unknown. Here, we show that beta-frequency communication between ventral hippocampus and basolateral amygdala also predicts anxiety-related behavior in mice, both on long timescales (~30 min) and immediately preceding behavioral choices. Genetically encoded voltage indicators reveal that this biomarker reflects synchronization between somatostatin interneurons across both structures. Indeed, synchrony between these neurons dynamically predicts approach-avoidance decisions, and optogenetically shifting the phase of synchronization by just 25 ms is sufficient to bidirectionally modulate anxiety-related behaviors. Thus, back-translation establishes a human biomarker as a causal determinant (not just predictor) of emotional state, revealing a novel mechanism whereby interregional synchronization that is frequency, phase, and cell type specific controls emotional processing.

INTRODUCTION

Anxiety normally acts as an instructive signal, promoting adaptive avoidance behaviors that help individuals navigate an uncertain world. However, in anxiety disorders, these signals become maladaptive, producing excessive and inappropriate avoidance. Collectively, anxiety disorders comprise the most common psychiatric disorder in the world and cause a high burden of disease.^{1,2} Identifying robust and valid biomarkers of psychiatric disorders is crucial for improving prevention, diagnosis, and treatment. Preclinical and clinical studies have identified limbic structures as key regulators of emotion.³ Recently, intracranial electroencephalogram (iEEG) recordings in humans with epilepsy revealed an electrophysiological biomarker that tracks real-time fluctuations in an individual's self-reported emotional state.⁴ This biomarker, based on beta-frequency coherence (β ; 13–30 Hz) between the amygdala (AMY) and hippocampus (HPC), advanced our understanding of how rapid timescale activity patterns within the human brain encode emotional states. In particular, this biomarker was conserved across most sub-

jects; could predict ~40% of the variance in scores on a “Immediate Mood Scaler” (IMS), which includes items related to both mood and anxiety; and tended to be valid in individuals with high baseline anxiety. However, as this biomarker was identified within human iEEG recordings, investigating its underlying cellular mechanisms and functional significance is challenging.

Rodent studies have investigated the roles of the AMY and HPC in anxiety-related behaviors. Ventral hippocampus (vHC) lesions reduce anxiety-related behaviors.^{5,6} Consistent with this, many vHC cells become active in anxiogenic environments, and opto- or chemogenetically modulating vHC outputs bidirectionally influences anxiety-dependent behaviors.^{7–12} The AMY is composed of several subnuclei, of which the basolateral amygdala (BLA) complex and central AMY have been implicated in anxiety processing.^{13,14} Projections from the BLA are thought to convey anxiety-related information to vHC.^{15–18} Although these studies highlight a role for vHC-BLA interactions in anxiety, little is known about the specific underlying neural activity patterns.

Preclinical studies typically start with behaviors presumed (based on face validity) to reflect disease-relevant emotional

states, identify associated patterns of neural activity, and presume that similar patterns will translate to humans. By contrast, here, we started with an established human biomarker (AMY-HPC β -coherence). Using homecage local field potential (LFP) recordings, we found that this human biomarker also predicts the emotional states of mice (as assayed by their subsequent anxiety-related avoidance behavior). This biomarker was specifically associated with the synchronization of somatostatin (SST+) interneurons within the BLA and vHC. The synchronization of these cell populations increases in anxiogenic environments and immediately preceding avoidance behaviors in the elevated plus maze (EPM). Artificially inducing or disrupting this synchronization, using 20 Hz stimulation that is in phase or shifted 25 ms out of phase, respectively, bidirectionally modulates these anxiety-related behaviors. Thus, this cross-species biomarker is a driver (not just readout) of emotional state. Furthermore, these results reveal a novel role for precise phase relationships between specific cell types distributed across limbic structures in creating network states that control emotional responses. Finally, BLA-SST+ neurons actually send long-range inhibitory connections to the vHC and appear to play a key role in this network state.

RESULTS

Hippocampus-amygdala β -coherence variance is a cross-species biomarker of negative emotion driven by brief β -coherence bursts

We sought to test whether the variance of β -coherence between the AMY and HPC, which tracks human mood and anxiety,⁴ also predicts rodent anxiety levels. In Kirkby et al., we measured this variance-based biomarker on multiple occasions and then determined how well the variation in this biomarker tracked changes in emotional state for an individual subject across those same time points. This requires repeated assessment of emotional state from a single subject. To do this in mice, we used the elevated zero maze (EZM). EZM performance is relatively stable over time and not subject to the “one-trial tolerance” phenomenon associated with the EPM.¹⁹ We recorded LFPs from BLA and vHC to compute the variance of β -coherence between these regions while each animal was in its homecage. Then, immediately following these 30-min-long home cage biomarker recordings, LFP recording was stopped, and we assayed EZM behavior for 5 min. We tested each mouse on 3 such sessions, with a 48-h inter-session interval (Figure 1A). Finally, in accordance with Kirkby et al., we normalized both biomarker and behavioral measurements on a within-subject basis (by computing Z scores based on each mouse’s three sessions). This enabled us to assess how well session-to-session changes in the biomarker for a mouse tracked session-to-session changes in its behavior. This is critical because we are studying a biomarker that explains within-subject changes in emotional state, rather than between-subject variation in trait anxiety levels. Moreover, normalizing measurements within each subject and then pooling data across subjects makes it possible to determine how well this biomarker for individual variation generalizes across subjects.

Consistent with human data, there was a strong negative relationship between the variance of vHC-BLA β -coherence measured in a mouse’s homecage and subsequent open-arm exploration in

the EZM (i.e., increases in the biomarker indicate higher baseline anxiety; Figure 1B). This correlation with behavior was not observed for the mean β -coherence or the mean or variance of coherence in other frequency bands (Figure S1).

The variance of coherence was calculated using a 60 s window, suggesting that high variance may reflect transient periods of high coherence superimposed on a low coherence baseline. Indeed, time-frequency plots for vHC-BLA β -coherence reveal brief periods of high β -coherence, termed here “ β -coherence bursts” (Figure 1C). The number of these bursts strongly predicts the overall level of vHC-BLA β -coherence within each 10 s window (Figure 1D) and also had a significant positive relationship with β -coherence variance (Figure 1E), suggesting that β -coherence bursts are a major contributor to this biomarker. Supporting this, we observed a significant negative relationship between β -coherence burst number during pre-test homecage recordings and EZM open-arm time (Figure 1F). Comparing periods of high β -coherence variance with those of low β -coherence variance reveals that high variance is associated with increases in the number, duration, and bandwidth of β -coherence bursts (Figure S1).

Hippocampus-amygdala β -coherence bursts contribute to a human mood biomarker

The contribution of β -coherence bursts to β -coherence variance in rodents suggests that bursts may play an analogous role in human iEEG recordings. Using the originally published dataset from patients with epilepsy, we found periods of brief, high AMY-HPC β -coherence within human iEEG recordings (Figure 2A). In human data, the number of β -coherence bursts had a strong, positive correlation with the variance in AMY-HPC β -coherence biomarker (Figure 2B). Additionally, β -coherence burst occurrence was strongly negatively correlated with subjective ratings of mood and anxiety (IMS; Figure 2C). Correspondingly, recordings during negative emotional states (high IMS) contained significantly more β -coherence bursts than during positive emotional states (low IMS) (Figure 2D). Notably, in these patients, IMS score was previously shown to correlate with common rating scales for depression, anxiety, and rumination.⁴ The preceding findings show that the variance of HPC-AMY β -coherence, which is a biomarker for emotional state in both humans and mice, is driven in both species (at least in part) by β -coherence bursts that occur on much faster timescales.

β -Coherence bursts predict anxiety-related behaviors on fast timescales

Whereas coherence variance is calculated over minutes, β -coherence bursts occur on subsecond timescales, suggesting they may play a role in anxiety-related behaviors on rapid, behaviorally relevant timescales. To assess this, we used the EPM, which contains a center zone where mice decide to either approach an exposed and brightly lit open arm (open runs; green) or engage in anxiety-related avoidance by entering a closed arm (closed runs; black). During EPM behavior, we recorded vHC and BLA LFP signals (Figure 3A). We observed time-locked increases in β -coherence burst probability immediately prior to avoidance. This time-locked increase was absent when animals approached (rather than avoided) the open arm (Figure 3B). Thus, the frequency with which

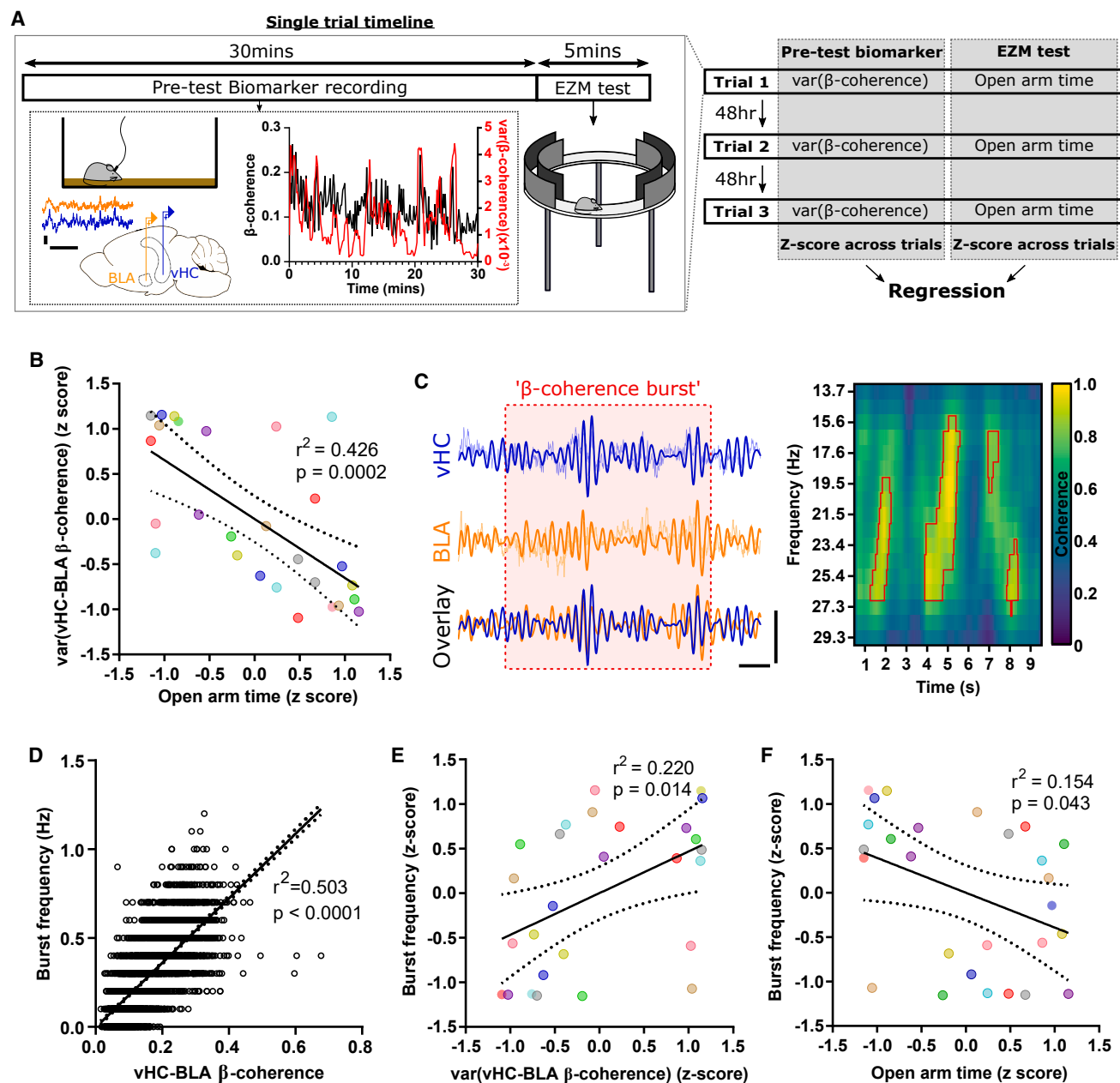


Figure 1. vHC-BLA variance of β -coherence is predictive of anxiety levels and is driven by transient high coherence bursts

(A) Experimental design. Left: each session began with 30 min home cage LFP recordings from BLA and vHC electrodes to calculate vHC-BLA β -coherence (black) and coherence variance (red). Mice were then immediately tested for anxiety-like behavior in the EZM. Right: each mouse was tested in 3 sessions with a 48 h inter-session interval. For each mouse, we used the β -coherence variance and EZM open-arm time from 3 sessions to calculate Z scores. Regressing these Z-scored measurements ($n = 27$ sessions from 9 mice) revealed their co-variation within a mouse and across days.

(B) Regression of Z-scored EZM open-arm time vs. Z-scored variance of vHC-BLA β -coherence home cage LFP recordings. Each point is data from one subject during one session. Different colors denote different subjects.

(C) Left: example vHC (raw: light blue; 13–30 Hz: blue) and BLA (raw: light orange; 13–30 Hz: orange) LFPs during a β -coherence burst (scale bars: 1 AU, 200 ms). Right: example coherogram displaying vHC-BLA β -coherence bursts (red outline).

(D) Regression of the frequency of β -coherence bursts (i.e., how often bursts occur) against mean vHC-BLA β -coherence.

(E) Regression of vHC-BLA β -coherence variance against β -coherence burst frequency from the same session. Both measures were Z scored within each subject.

(F) Regression of open-arm time against β -coherence burst frequency from the same session. Both measures were Z scored within each subject.

All panels: dotted boundaries represent 95% confidence estimates.

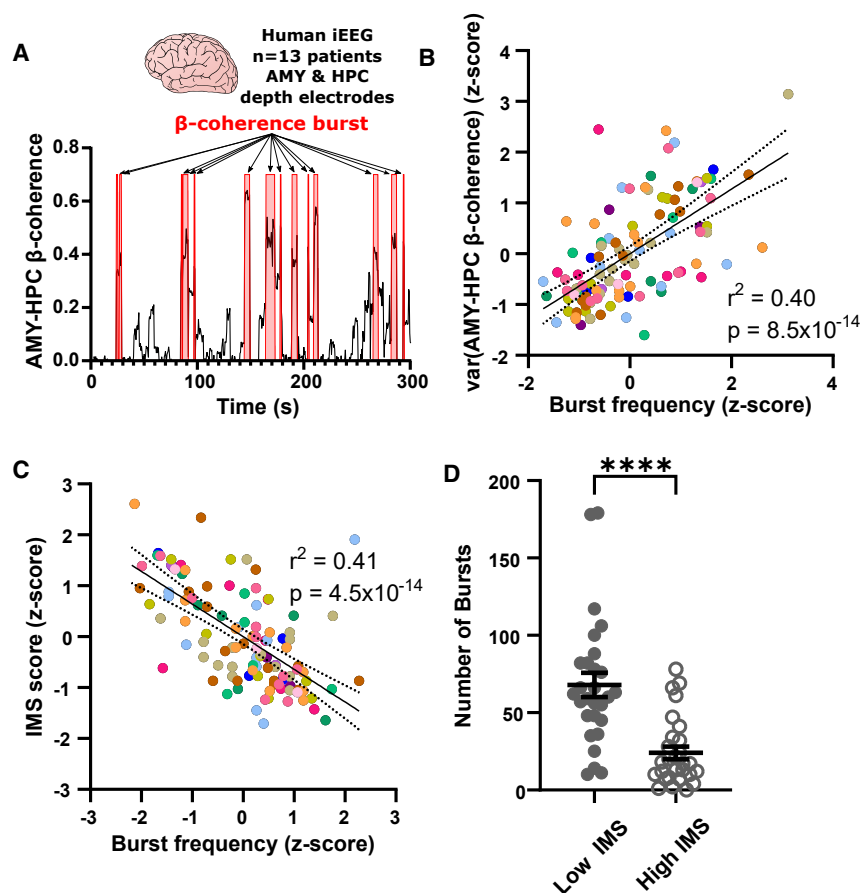


Figure 2. β -coherence bursts drive human anxiety biomarker and predict real-time mood fluctuations

(A) Example trace of AMY-HPC β -coherence (black) in human iEEG recordings from epileptic patients, with β -coherence bursts (red).

(B) Regression of AMY-HPC β -coherence variance against the β -coherence burst frequency in iEEG recordings. Both measures Z scored within each subject. Each point is data from one subject during one session. Different colors denote different subjects.

(C) Regression of IMS score against β -coherence burst frequency. Both measures Z scored within each subject.

(D) β -coherence burst frequency was significantly increased during periods of low IMS score (lower quartile) compared with high IMS (upper quartile) periods ($p < 0.0001$, $n = 28$; Mann-Whitney test). Points represent values from individual sessions with mean $\pm$ SEM.

All panels: dotted boundaries represent 95% confidence estimate.

β -coherence bursts occurs was significantly higher during the 3 s preceding closed-arm entry than during the period preceding open-arm entry (Figure 3C). Although average locomotion speed was higher during closed runs than during open runs, the average speed in the center zone (before mice entered the closed or open arms) did not differ (Figure S2). Furthermore, there was no relationship between locomotion speed and β -coherence burst occurrence for either run type (Figure S2). Additionally, analysis of “burst-like” periods of coherence in the low gamma band (30–50 Hz) did not reveal significant differences that predicted approach-avoidance decisions (Figure S2).

vHC and BLA SST+ interneurons synchronize during β -coherence bursts

GABAergic interneurons have been implicated in neuronal oscillations.²⁰ We, therefore, tested whether β -coherence bursts were accompanied by the synchronization of specific GABAergic cell types within the BLA and vHC using *trans*-membrane electrical measurements performed optically (TEMPO). TEMPO uses bulk fluorescence from the genetically encoded voltage indicator (GEVI) Ace2N-4AA-mNeon (mNeon) to monitor voltage signals in specific cell types.²¹ We injected AAV-CAG-DIO-Ace2N-4AA-mNeon and AAV-Synapsin-tdTomato unilaterally into both the BLA and the vHC of SST-Cre or PV-Cre mice to drive Cre-dependent expression of mNeon in SST+ or parvalbumin (PV+) interneu-

rons, respectively, along with the nonspecific expression of a reference fluorophore (tdTomato). We then implanted optrodes to measure mNeon and tdTomato fluorescence and LFPs (Figure 4A). GEVI fluorescence is contaminated by both movement and hemodynamic artifacts.^{21,22} To minimize the influence of these artifacts, after filtering signals in the frequency band of interest, we used robust linear regression to fit each tdTomato signal to the corresponding mNeon signal and then subtracted the rescaled tdTomato signal to partially correct the mNeon signal (Figure 4B). To assess the relationship between interneuron synchrony and vHC-BLA β -coherence bursts, the onset of each β -coherence burst that showed high correlation (>0.5) between BLA and vHC-LFP recordings was time locked to simultaneously recorded mNeon signals. To measure β -synchrony within these interneuron populations, we calculated the cross-correlation between vHC and BLA mNeon signals (which had been filtered in the β -frequency band) during each β -coherence burst. This revealed strong β -frequency correlation between voltage signals from BLA and vHC-SST+ populations during β -coherence bursts. Furthermore, this was not observed in PV-Cre mice (Figure 4C).

To confirm that this increase in β -frequency correlation reflects synchronization (phase locking) between BLA and vHC-SST+ neurons, we computed the phase-locking value (PLV) between mNeon signals from these populations. In recordings from SST-Cre mice, PLVs were significantly greater during β -coherence bursts than during random time points lacking β -coherence bursts (Figure 4D). Notably, this was not the case when we calculated PLVs in PV+ neurons (Figure 4E). There also was no correlation between each mNeon signal and the tdTomato signal from the other region, confirming that this correlation was not artifact driven (Figure S3). Interestingly, the peak cross-correlation

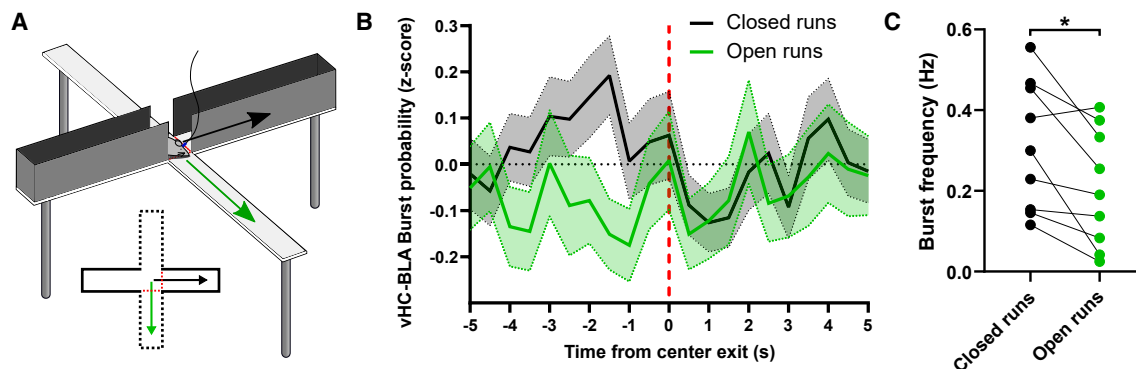


Figure 3. vHC-BLA β -coherence bursts predict approach-avoidance decisions

(A) Schematic: LFP signals were analyzed during EPM, time locked to open-arm (green) or closed-arm (black) entries.

(B) vHC-BLA β -coherence burst probability as animals enter closed (black) or open arms (green).

(C) vHC-BLA β -coherence burst frequency was significantly higher during the 3 s prior to closed runs than during the period prior to open runs ($p = 0.0122$, $n = 9$ mice; paired t test). Points represent values from individual mice.

occurred when BLA-SST+ mNeon signals preceded vHC-SST+ mNeon signals by approximately 90° , suggesting that β -frequency activity in BLA-SST+ neurons leads vHC-SST+ neurons.

Plotting the in-phase correlation between BLA-SST+ and vHC-SST+ interneurons around the onset of β -coherence bursts revealed that vHC-BLA-SST+ neuron β -synchrony increases immediately following burst onset (Figure 4F). For each time point, we compared the correlation in real data with that of 100 shuffled datasets in which one mNeon signal was randomly time shifted (Figure 4G).

Interestingly, we analyzed the correlation between θ -frequency (4–8 Hz) components of BLA and vHC-SST+ neuron mNeon signals and found that this correlation increases 500 ms prior to β -coherence burst onset and during the first 500 ms of the burst (Figure S3). No significant increases in γ -correlation (30–50 Hz) occurred around β -coherence bursts for either SST-Cre or PV-Cre mice (Figure S3).

vHC and BLA-SST+ interneuron β -synchrony increases during anxiety-related behavior

Given this relationship between vHC-BLA-SST+ β -synchrony (calculated as described above) and the LFP β -coherence bursts that precede anxiety-related avoidance, we tested whether SST+ interneuron β -synchrony was similarly associated with anxiety-related behaviors. SST+ interneuron β -synchrony was higher in the EPM than in the homecage (Figure 5A). This effect was driven by increased β -synchrony within the EPM center zone (Figure 5B). Neither an overall increase in β -synchrony in the EPM (Figure 5C) nor any zone-specific changes (Figure 5D) occurred for PV+ interneurons. These changes were also specific for β -frequencies (Figure S4).

In LFP recordings, β -coherence bursts specifically precede open-arm avoidance (Figure 3B). Therefore, we examined whether increased SST+ interneuron β -synchrony in the EPM center reflects a relationship between β -synchrony and specific avoidance behaviors. Signals time locked to center zone exit revealed increased SST+ interneuron β -synchrony just prior to open-arm avoidance (i.e., closed-arm entries), but not open-arm

approaches (Figure 5F). The percentage of time points for which the β -synchrony (i.e., the correlation between β -band-filtered mNeon signals from SST+ interneurons) calculated from real data exceeded the 95th percentile of values based on shuffled data also increased during the 5 s preceding open-arm avoidance (Figure 5G) and was significantly elevated relative to the equivalent period preceding open-arm approaches (Figure 5H). These effects were not observed in PV+ interneurons (Figures 5I–5L) or in mNeon signals filtered from 4 to 8 or 30 to 50 Hz (Figure S4).

Similar to this increase in SST+ interneuron β -synchrony preceding open-arm avoidance, SST+ interneuron β -synchrony also increased prior to the termination of open-arm “probes”, during which an animal’s head enters the open arm while its body remains in the center zone (Figures 5M–5P). The fraction of time points for which the 13–30 Hz mNeon correlations exceeded the 95th percentile of shuffled data also increased prior to the termination of open-arm probes for SST+ interneurons (Figure 5O). Again, this was not observed for PV+ interneurons. Correspondingly, for the 1.5 s preceding the termination of open-arm probes, the fraction of time points exceeding the 95th percentile based on shuffled data was significantly greater in SST-Cre mice than in PV-Cre mice (Figure 5P). Thus, increased SST+ interneuron β -synchrony precedes the end of risk-assessment behaviors, i.e., when mice either transit from the center zone to a closed arm or terminate open-arm probes.

Optogenetic induction or disruption of vHC-BLA-SST+ interneuron β -synchrony bidirectionally modulates anxiety-related behaviors

Having established that the β -synchrony of SST+ interneurons across the vHC and BLA is an anxiety-related biomarker that precedes the termination of risk-assessment behaviors, we tested whether manipulating this pattern of activity would causally influence these behaviors. Bilateral fiber optics were implanted in the BLA and vHC of both SST-Cre and PV-Cre mice injected with Cre-dependent channelrhodopsin (AAV-DIO-ChR2-eYFP) (Figure 6A). As increases in β -synchrony occurred specifically within the EPM center zone, we delivered 473 nm

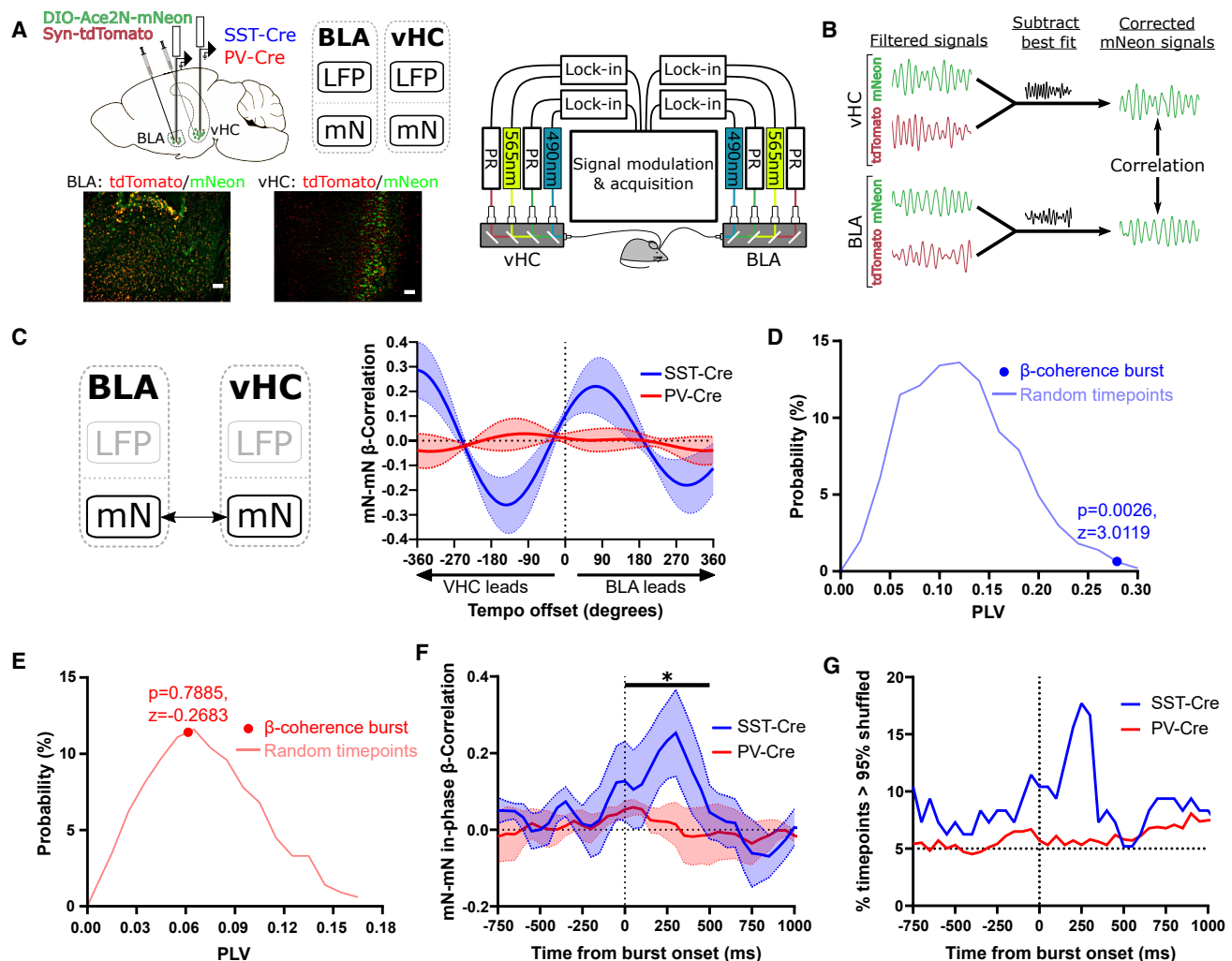


Figure 4. vHC and BLA-SST+ interneurons synchronize during vHC-BLA β -coherence bursts

(A) Left: SST-Cre or PV-Cre mice had unilateral viral injections plus optrode implants in the BLA and vHC. Scale bars: 50 μ m. Right: schematic for dual-site TEMPO.

(B) Overview of TEMPO signal cleaning and analysis.

(C) Cross-correlation of BLA mNeon and vHC mNeon β -frequency signals during β -coherence bursts in SST-Cre (blue) and PV-Cre (red) mice.

(D) Actual phase-locking values for BLA and vHC-SST+ mNeon β -frequency signals during β -coherence bursts compared with distribution from 1,000 time-shuffled datasets ($p = 0.0026$, $Z = 3.0119$, $n = 6$ mice, Z test).

(E) Actual phase-locking values for BLA and vHC-PV+ mNeon β -frequency signals during β -coherence bursts compared with distribution from 1,000 time-shuffled datasets ($p = 0.7885$, $Z = -0.2683$, $n = 7$ mice, Z test).

(F) Peak in-phase correlation was significantly higher 0–500 ms post β -coherence burst onset in SST-Cre mice (blue) than in PV-Cre mice (red) ($p = 0.047$, SST-Cre $n = 6$, PV-Cre $n = 7$; unpaired t test).

(G) The fraction of SST+ (blue) and PV+ (red) mNeon correlation values above the 95th percentiles of values derived from time-shuffled data.

light whenever mice were in a “stimulation zone” comprising the center zone and proximal thirds of open and closed arms. Behavior was compared between three 3-min-long epochs: no stimulation, an “in-phase” period (same stimulation was delivered bilaterally to both the BLA and vHC), and an “out-of-phase” period (bilateral stimulation was delivered that was 180° out of phase between the BLA and vHC) (Figure 6B). In-phase optogenetic stimulation promotes normal β -synchrony by delivering the same rhythmic stimulation to both structures, whereas out-of-phase stimulation is designed to disrupt this

condition using misaligned light pulses. Importantly, in-phase and out-of-phase refers to the phase relationship between interneurons in the BLA and vHC (not to relationships between the pulses and endogenous LFPs).

Disrupting SST+ interneuron β -synchrony using out-of-phase 20 Hz stimulation increased the amount of time in the center zone of the EPM relative to both no stimulation and in-phase stimulation (Figure 6C). This was not observed when stimulation was delivered at 8 Hz or in PV-Cre mice (Figure 6C). This increased center time caused by out-of-phase 20 Hz stimulation occurs

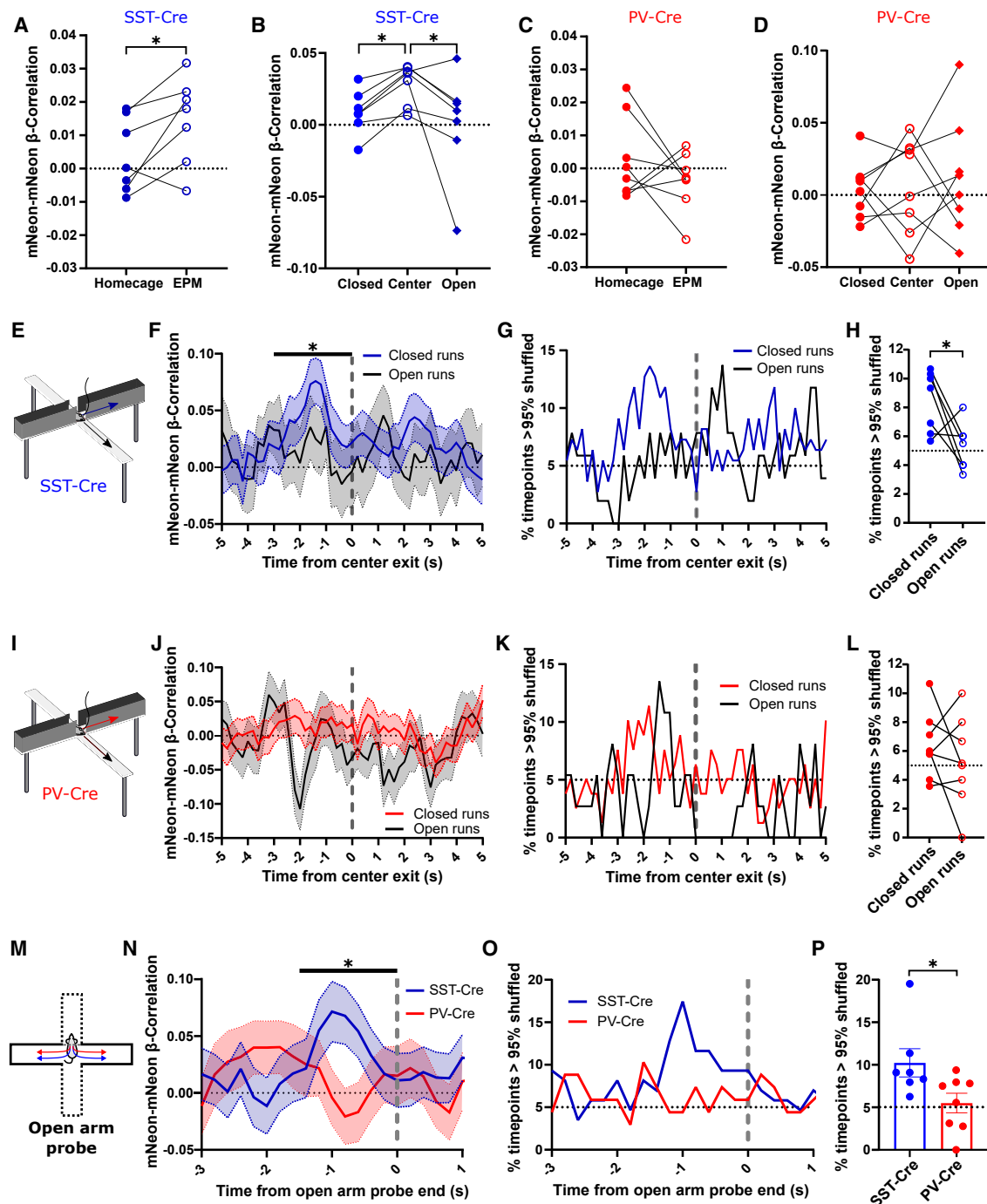


Figure 5. vHC-BLA-SST+ interneuron β -synchrony predicts EPM behavior

(A) SST+ interneuron β -synchrony was significantly higher in the EPM than in the home cage ($p = 0.038$, $n = 7$ mice; paired t test). Points represent values from individual mice.

(B) SST+ interneuron β -synchrony significantly increases in the EPM center zone relative to open and closed arms ($p = 0.0207$, $n = 7$ mice; Friedman test with Dunn's correction; closed vs. center: post hoc $p = 0.0485$, closed vs. open: post hoc $p > 0.99$, center vs. open: post hoc $p = 0.0485$). Points represent values from individual mice.

(C) PV+ interneuron β -synchrony in home cage vs. EPM ($p = 0.2985$, $n = 8$ mice; paired t test). Points represent values from individual mice.

(D) PV+ interneuron β -synchrony was not influenced by location within the EPM ($p = 0.7829$, $n = 8$ mice; repeated measures one-way ANOVA). Points represent values from individual mice.

(E) 13–30 Hz SST+ interneuron mNeon signals were analyzed during EPM, time locked to open-arm (black) or closed-arm (blue) entries.

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at the expense of closed-arm time (Figure 6E). Furthermore, within the center zone, out-of-phase 20 Hz stimulation increased the time *SST-Cre* mice spent engaged in open-arm probes (Figure 6F). Again, this effect was not observed with 8 Hz stimulation in *SST-Cre* mice or with 20 Hz stimulation in *PV-Cre* mice (Figure 6F). Thus, optogenetically disrupting the increases in *SST+* interneuron β -synchrony (which normally occur when mice engage in avoidance behaviors and/or terminate risk assessment) has the effects of decreasing avoidance and increasing risk-assessment behavior.

In contrast, in-phase 20 Hz stimulation specifically increases closed-arm runs in *SST-Cre* mice (Figure 6G). The effect of in-phase stimulation on closed-arm runs did not reach significance when comparing all three groups separately (group $\times$ stim session: $p = 0.083$) but was significant for a single comparison between *SST-Cre* mice receiving 20 Hz stimulation and a combined control group (*SST-Cre* 8 Hz and *PV-Cre* 20 Hz) (group $\times$ stim session: $p = 0.045$). Neither stimulation pattern had any effect on the number of open runs in any of groups (Figure 6H). Thus, artificially inducing β -synchrony in *SST+* interneurons can promote avoidance behavior (closed-arm runs) but is not, on its own, sufficient to disrupt open-arm approaches.

To confirm the specificity of these effects for anxiety-related behaviors, we tested *SST-Cre* mice using a cross maze, a modified EPM containing 4 closed arms. Neither stimulation pattern influenced center time in the cross maze (Figure S5). Additionally, neither influenced the probing of cross maze arms that are geographically equivalent to either the EPM open or closed arms (Figure S5). This shows that the observed effects on EPM behavior reflect changes in anxiety-dependent approach-avoidance conflicts and are not just artifacts related to maze geometry or nonspecific stimulation effects.

BLA-SST+ neurons increase phase locking with local and long-range circuits during β -coherence bursts

Simultaneously recording LFPs and GEVI signals can reveal relationships between specific cell types and population/circuit-

level activity. The cross-correlation between β -frequency activity in BLA-SST+ voltage signals and BLA LFPs, calculated during ongoing β -coherence bursts, shows high correlation around specific phases (Figure 7A). Indeed, the PLV between these signals significantly increases during β -coherence bursts (Figure 7B). This was not observed for BLA-PV+ neurons (Figure 7C). A similar pattern of cross-correlation and elevated PLV occurs for β -frequency activity in BLA-SST+ neurons and vHC LFPs (Figures 7D and 7E). Again, this was not observed for BLA-PV+ neurons (Figure 7F). Additionally, although LFP β -power increased in both regions during β -coherence bursts, the β -power of SST+ neuron voltage signals did not (Figure S6). This suggests that β -coherence bursts reflect increased phase synchronization between BLA-SST+ neurons and network activity, not an increase in the amplitude of β -frequency activity in BLA-SST+ neurons. In contrast to BLA-SST+ neurons, we did not observe significant β -synchrony between vHC-SST+ neurons and vHC or BLA LFPs (Figure S6). Interestingly, significant β -synchrony between vHC-PV+ cells and vHC LFP was observed during β -coherence bursts (Figure S6). These data, along with the observation that β -activity in BLA-SST+ neurons precedes that in vHC-SST+ neurons, suggest that BLA-SST+ cells may play a key role in entraining both local circuit and distributed network activity during β -coherence bursts.

Role of BLA-SST+ cells in burst generation

Projections from the AMY are thought to provide anxiety-related information to vHC circuits.^{15–18} The presence of inhibitory long-range projections from the BLA to the vHC¹⁶ and β -synchrony between BLA-SST+ interneurons and BLA LFP, vHC LFP, and vHC-SST+ voltage signals suggests that BLA-SST+ interneurons may play a key role in β -coherence bursts. Therefore, we used optogenetics to causally test whether BLA-SST+ interneurons help generate β -coherence bursts. An optrode was implanted in the BLA of *SST-Cre* mice along with Cre-dependent expression of either halorhodopsin or a control fluorophore; an additional LFP electrode was implanted in vHC (Figure 8A).

(F) SST+ interneuron 13–30 Hz mNeon peak correlation was significantly higher in the 3 s preceding closed (blue) runs than during the period preceding open (black) runs ($p = 0.0228$, closed runs $n = 110$, open runs $n = 51$; unpaired t test).

(G) The fraction of SST+ interneuron mNeon correlation values $>95^{\text{th}}$ percentile of values derived from time-shuffled mNeon signals, during open-arm (black) or closed-arm (blue) runs.

(H) SST+ interneurons β -synchrony was significantly greater during the 5 s preceding closed-arm entries than during the period preceding open-arm entries ($p = 0.039$, $n = 7$ mice; paired t test). Points represent values from individual mice.

(I) 13–30 Hz PV+ interneuron mNeon signals were analyzed during EPM, time locked to open-arm (black) or closed-arm (red) entries.

(J) PV+ interneuron 13–30 Hz mNeon peak correlation was not significantly different in the 3 s preceding closed (red) runs compared with the period preceding open (black) runs ($p = 0.4686$, closed runs $n = 79$, open runs $n = 37$; unpaired t test).

(K) The fraction of PV+ interneuron mNeon correlation values $>95^{\text{th}}$ percentile of values derived from time-shuffled mNeon signals, during open-arm (black) or closed-arm (red) runs.

(L) PV+ interneuron β -synchrony during the 5 s preceding closed- or open-arm entries ($p = 0.3557$, $n = 8$ mice; paired t test). Points represent values from individual mice.

(M) Schematic of open-arm probe.

(N) SST+ interneuron (blue) and PV+ interneuron (red) 13–30 Hz mNeon correlation values aligned to the termination of open-arm probes. During the 1.5 s prior to the termination of each open-arm probe, the peak correlation was significantly higher in *SST-Cre* mice than in *PV-Cre* mice ($p = 0.0424$, *SST-Cre* probes $n = 86$, *PV-Cre* probes $n = 68$; unpaired t test).

(O) The fraction of SST+ interneuron (blue) and PV+ interneuron (red) mNeon correlation values $>95^{\text{th}}$ percentile of values derived from time-shuffled mNeon signals.

(P) Based on the metric plotted in (O) (fraction of time points $>95^{\text{th}}$ percentile of values derived from shuffled data), β -synchrony was significantly greater for SST+ interneurons than for PV+ interneurons during the 1.5 s preceding the termination of open-arm probes ($p = 0.027$, *SST-Cre* $n = 7$, *PV-Cre* $n = 8$; Mann-Whitney test). Points represent values from individual mice.

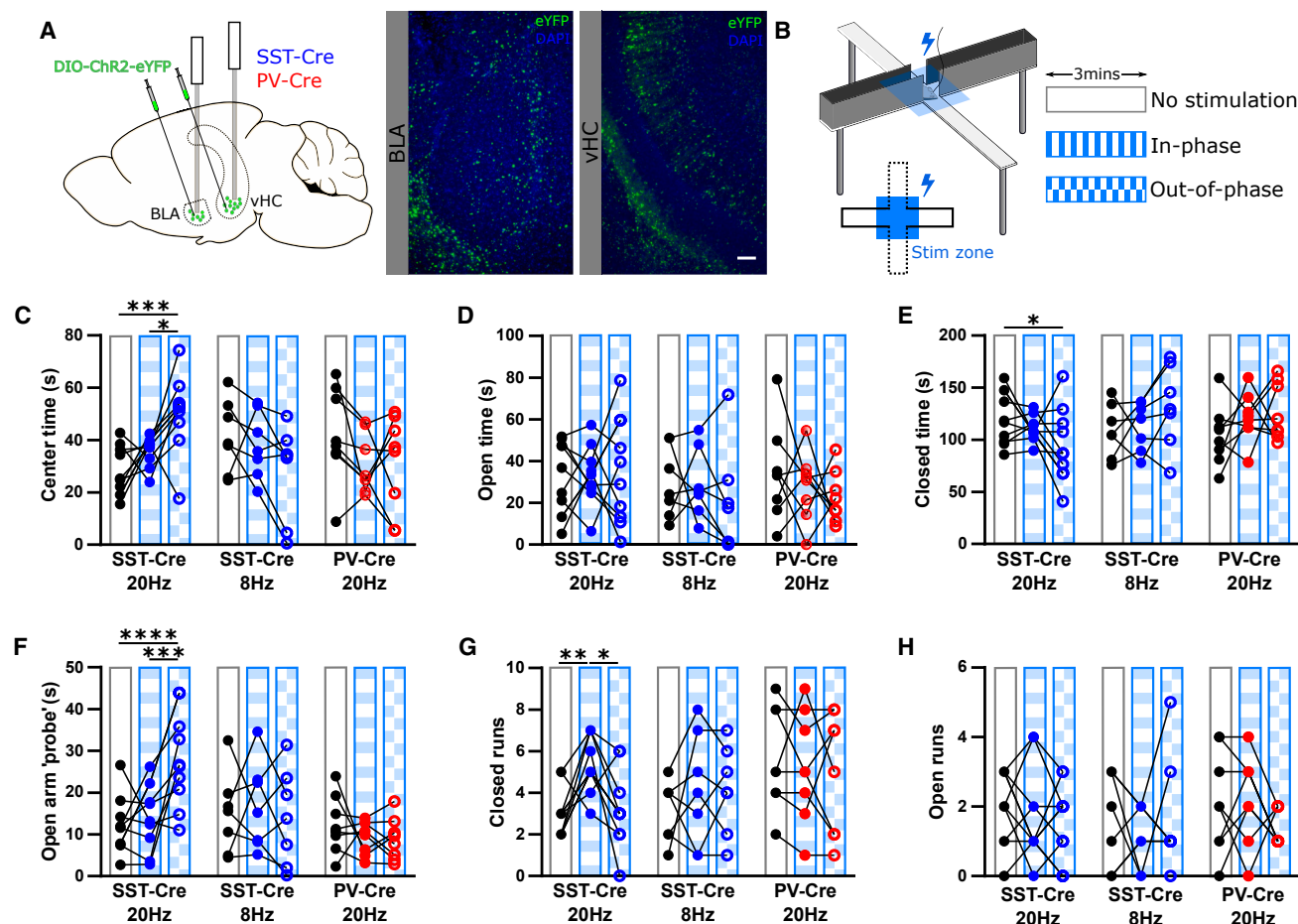


Figure 6. Modulating vHC-BLA-SST+ interneuron β -synchrony bidirectionally influences approach-avoidance behaviors

(A) Left: experimental design showing the location of optic fiber implants in the BLA and vHC. Right: images of the BLA and vHC illustrating ChR2-eYFP expression. Scale bars: 100 μ m.

(B) The stimulation paradigm comprised three 3 min epochs, during which there was no stimulation, in-phase stimulation, or out-of-phase stimulation.

(C) 20 Hz out-of-phase stimulation in SST-Cre mice significantly increased center-zone time (two-way ANOVA [group $\times$ stim session]; interaction: $F_{(4,44)} = 7.908$, $p < 0.0001$; SST-Cre 20 Hz post hoc with Tukey's multiple comparison test: no stim vs. in-phase: $p = 0.187$; no stim vs. out-of-phase: $p < 0.001$; in-phase vs. out-of-phase: $p = 0.037$). Points represent values from individual mice.

(D) In-phase or out-of-phase stimulation at 8 or 20 Hz had no effect on open-arm time in any group (two-way ANOVA [group $\times$ stim session]; interaction: $F_{(4,44)} = 0.5759$, $p = 0.682$). Points represent values from individual mice.

(E) 20 Hz out-of-phase stimulation in SST-Cre mice significantly decreased closed-arm time (two-way ANOVA [group $\times$ stim session]; interaction: $F_{(4,44)} = 3.193$, $p = 0.022$; SST-Cre 20 Hz post hoc Tukey's multiple comparison test: no stim vs. in-phase: $p = 0.620$; no stim vs. out-of-phase: $p = 0.049$; in-phase vs. out-of-phase: $p = 0.303$). Points represent values from individual mice.

(F) 20 Hz out-of-phase stimulation in SST-Cre mice significantly increased open-arm probing (two-way ANOVA [group $\times$ stim session]; interaction: $F_{(4,44)} = 5.225$, $p = 0.0016$; SST-Cre 20 Hz post hoc Tukey's multiple comparison test: no stim vs. in-phase: $p = 0.706$; no stim vs. out-of-phase: $p < 0.0001$; in-phase vs. out-of-phase: $p < 0.001$). Points represent values from individual mice.

(G) 20 Hz in-phase stimulation in SST-Cre mice significantly increased the number of closed-arm runs (two-way ANOVA [group $\times$ stim session]; interaction: $F_{(4,44)} = 2.211$, $p = 0.0447$; SST-Cre 20 Hz post hoc Tukey's multiple comparison test: no stim vs. in-phase: $p = 0.0057$; no stim vs. out-of-phase: $p = 0.99$; in-phase vs. out-of-phase: $p = 0.0123$). Points represent values from individual mice.

(H) Neither in-phase nor out-of-phase stimulation at 8 or 20 Hz had any effect on open-arm runs in any group (two-way ANOVA [group $\times$ stim session]; interaction: $F_{(4,44)} = 0.868$, $p = 0.491$). Points represent values from individual mice.

Optogenetically inhibiting BLA-SST+ interneurons triggered a decrease in β -coherence bursts that was not observed in non-opsin controls (Figures 8B and 8C).

To further validate our hypothesized link between changes in β -coherence bursts and anxiety-related avoidance, we inhibited BLA-SST+ interneurons during EPM behavior. Inhibiting BLA-

SST+ interneurons significantly increased open-arm exploration (Figure S7), as expected, based on our previous finding that optogenetically inhibiting this population suppresses β -coherence bursts. However, this increase in open-arm exploration was the result of a prolongation of open-arm visits rather than an increase in the probability of open-arm entry (Figure S7). Furthermore,

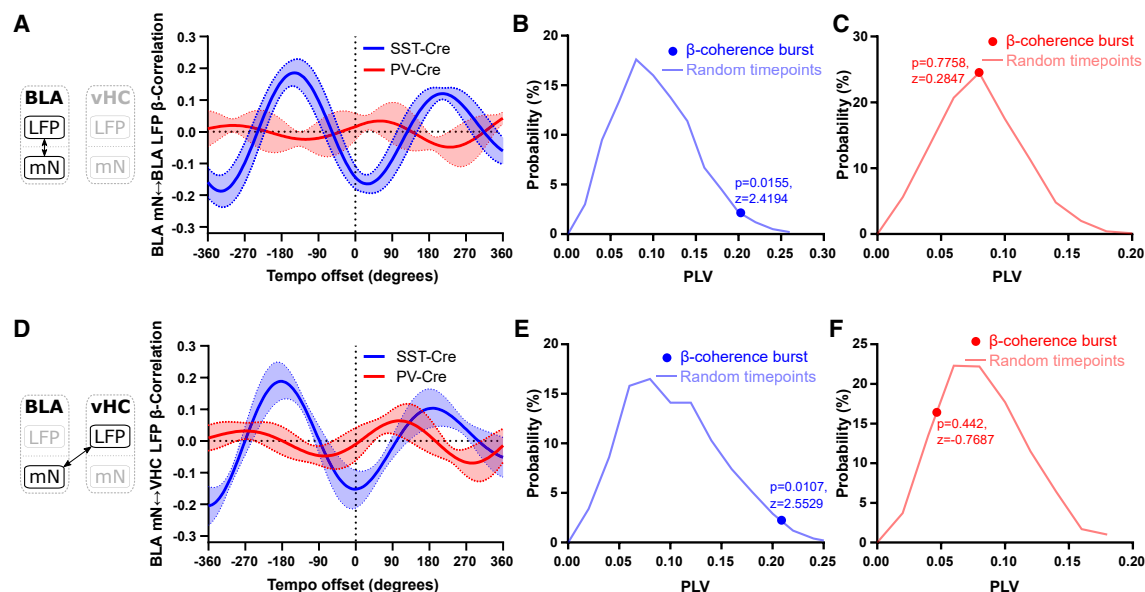


Figure 7. BLA-SST+ interneurons synchronize with BLA and vHC LFPs during β -coherence bursts

(A) Cross-correlation between BLA mN and BLA LFP β -frequency signals during β -coherence bursts in SST-Cre (blue) and PV-Cre (red) mice. (B) Distribution of β -frequency phase-locking values (PLVs) calculated from 1,000 time-shuffled pairs of BLA-SST+ neuron mN signals and BLA LFPs, compared with the actual PLV during β -coherence bursts ($p = 0.0115$, $Z = 2.4149$, $n = 6$ mice, Z test). (C) Distribution of β -frequency phase-locking values (PLVs) calculated from 1,000 time-shuffled pairs of BLA-PV+ neuron mN signals and BLA LFPs, compared with the actual PLV during β -coherence bursts ($p = 0.7758$, $Z = 0.2847$, $n = 6$ mice, Z test). (D) Cross-correlation between BLA mN and vHC LFP β -frequency signals during β -coherence bursts in SST-Cre (blue) and PV-Cre (red) mice. (E) Distribution of β -frequency phase-locking values (PLVs) calculated from 1,000 time-shuffled pairs of BLA-SST+ neuron mN signals and vHC LFPs, compared with the actual PLV during β -coherence bursts ($p = 0.0107$, $Z = 2.5529$, $n = 6$ mice, Z test). (F) Distribution of β -frequency phase-locking values (PLVs) calculated from 1,000 time-shuffled pairs of BLA-PV+ neuron mN signals and vHC LFPs, compared with the actual PLV during β -coherence bursts ($p = 0.442$, $Z = -0.7687$, $n = 7$ mice, Z test).

when we performed a real-time place preference assay using the optogenetic inhibition of BLA-SST+ interneurons, mice spent significantly more time in the zone where optogenetic inhibition was delivered (Figure S7). This confounds the interpretation of this EPM experiment, as increases in open-arm time may reflect a change in “location preference” rather than anxiety-related behavior per se.

An intriguing possibility suggested by Figure 1 is that β -coherence burst frequency may not only affect ongoing anxiety-related behaviors but also predict future behaviors that occur several minutes later. To test this, we inhibited BLA-SST+ interneurons for 30 min while animals explored their homecage. Then, the stimulation was stopped, and mice were tested in the EZM (Figure 8D). Homecage BLA-SST+ inhibition resulted in significantly greater open-arm exploration in eNpHR-expressing mice than in non-opsin controls (Figure 8E). This suggests that the prolonged suppression of β -coherence bursts can elicit sustained reductions in anxiety-related avoidance, which outlast the period of optogenetic inhibition.

The finding that BLA-SST+ interneurons seem to play a key role in initiating β -synchrony across BLA and vHC circuits suggests a potential role for long-range, direct inhibitory connections from BLA-SST+ neurons onto vHC targets, potentially including SST+ interneurons. To test this, we injected AAV-Cre-ON/Flp-OFF-ChR2-eYFP virus into the BLA of SST-Cre mice along with

a separate virus encoding Flpase into the vHC (Figure 8F). This strategy prevented potential cross-regional leak, i.e., the direct transduction of vHC cell bodies within the vHC due to low-efficiency spread of AAV injected into the BLA. Whole-cell patch-clamp recordings from vHC-SST+ neurons and optogenetic activation of BLA-SST+ inputs revealed opto-inhibitory postsynaptic currents (IPSCs) (~ 50 – 200 pA) that were insensitive to glutamatergic blockers but abolished by the GABA-A blocker gabazine (Figures 8F and 8G). This confirms the presence of robust monosynaptic connections from BLA-SST+ neurons to vHC-SST+ interneurons, which could synchronize both these SST+ populations and broader AMY-HPC circuits.

DISCUSSION

This study reveals that β -frequency communication between the AMY and HPC predicts emotional state in mice, back-translating previous findings from humans. There are four other major advances. First, in both humans and mice, the biomarker is driven by β -coherence bursts, transient periods of high β -coherence, which predict mouse approach-avoidance behaviors on time-scales of seconds. Second, GEVIs reveal that these bursts are associated with β -frequency synchronization between SST+ interneurons in the BLA and vHC and that the β -synchrony of these populations similarly increases during avoidance behaviors.

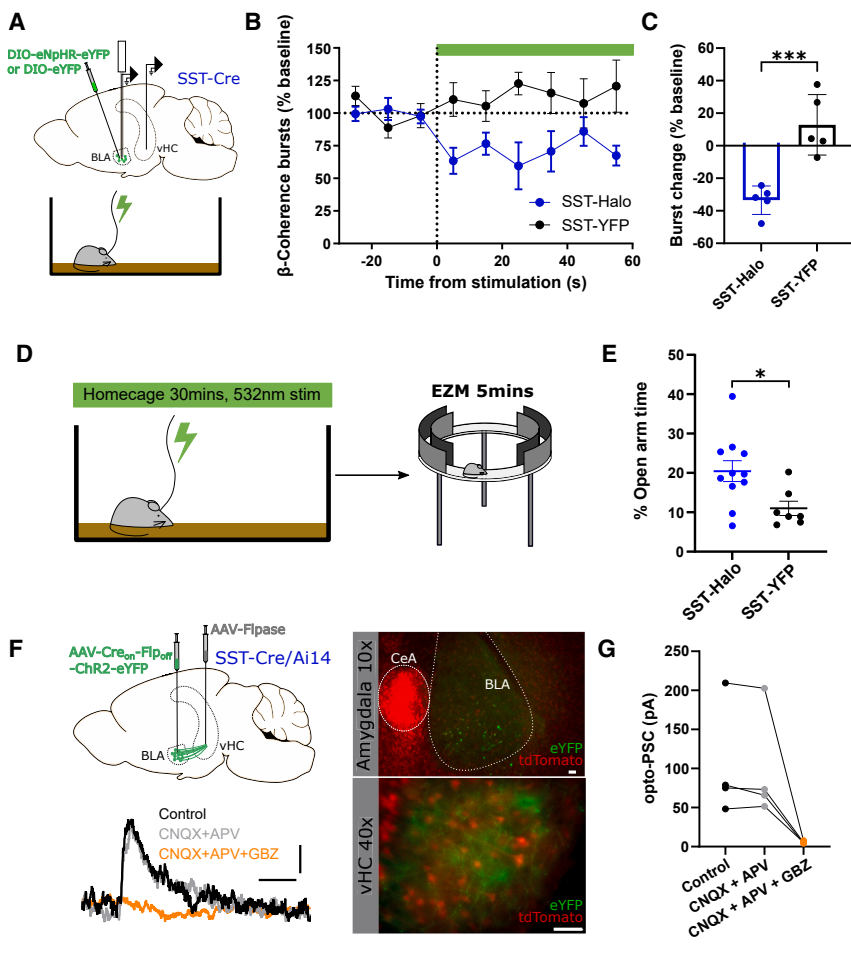


Figure 8. Inhibiting BLA-SST+ interneurons decreases β -coherence bursts and anxiety-related avoidance

(A) SST-Cre mice received unilateral BLA injections along with an optrode implant in the BLA and electrode implant in the vHC.

(B) Time course of how inhibiting BLA SST+ interneurons affects β -coherence bursts.

(C) β -coherence bursts were significantly suppressed by BLA-SST+ inhibition relative to non-opsin controls ($p = 0.001$, SST-Halo $n = 5$, SST-YFP $n = 5$, unpaired t test). Points represent values from individual mice.

(D) Stimulation paradigm: 532 nm stimulation was delivered in a 90 s on/off cycle for 30 min in the homecage. The stimulation was then stopped and followed by 5 min of EZM testing.

(E) Homecage BLA-SST+ inhibition significantly increased EZM open-arm exploration compared with non-opsin controls ($p = 0.0201$, SST-Halo $n = 11$, SST-YFP $n = 7$, unpaired t test). Points represent values from individual mice.

(F) Top left: experimental design. Bottom left: example whole-cell voltage clamp traces showing an optically evoked IPSC in a vHC-SST+ interneuron (black) and the effects of adding glutamatergic (gray) and then GABAergic blockers (orange). Scale bars: 25 ms, 25 pA. Top right: 10 $\times$ BLA image. Bottom right: 40 $\times$ vHC image showing eYFP+ fibers.

(G) Optically evoked post synaptic potentials in vHC-SST+ interneurons are abolished by gabazine (GBZ) but not 6-cyano-7-nitroquinoxaline-2,3-dione (CNQX) + d-2-amino-5-phosphonopentanoic acid (APV) ($p = 0.0417$, $n = 4$ cells; Friedman test with Dunn's correction; control vs. CNQX + APV: post hoc $p = 0.959$, control vs. CNQX + APV + GBZ: post hoc $p = 0.0267$). Points represent values from individual cells.

Third, artificially inducing or disrupting synchrony in these SST+ populations during approach-avoidance conflicts bidirectionally influences these behaviors in a phase- and frequency-dependent manner. Finally, BLA-SST+ neurons appear to play a particularly key role in this communication mode, potentially through their GABAergic projections to vHC-SST+ interneurons.

Precise phase relationships define distinct anxiety-related network states

Studies of circuits mediating emotional behaviors largely focus on how changes in the activity of specific cell types promote or suppress specific behaviors. An alternative has been identifying engrams, subsets of neurons that are activated during specific behaviors, e.g., freezing. This study reveals a novel and complementary dimension, whereby the synchronization of specific cell populations on much faster timescales (~ 10 –50 ms) and across brain regions defines network interactions that serve as markers of emotional state and causally influence approach-avoidance behaviors. Previous studies have linked hippocampal-prefrontal theta-frequency synchronization to anxiety-related behavior.^{8,23–27} These studies²⁷ indicated that anxiety-related input from vHC to prefrontal cortex is normally modulated at theta frequency, such that stimulation at this frequency is best for facil-

itating the transmission of this input. Here, we find that two patterns of stimulation that are both delivered at the same frequency can nevertheless elicit opposing effects on approach-avoidance behaviors in a phase-dependent manner. This further establishes that it is the precise phase relationship between neurons in different brain structures, not just the strength of a rhythmic input or activity level within a specific circuit, that defines distinct network states.

All of the effects we found, using voltage indicators and optogenetics, were specific to cell type, frequency, and phase relationship. These powerful controls confirm that these phenomena reflect the rhythmic synchronization of specific cell types, not just overall increases in neuronal activity or non-neuronal artifacts. Of note, the number of transfected cells was approximately 2-fold higher in the vHC of SST-Cre mice than in that of PV-Cre mice (Figure S5). Although we cannot fully rule out that higher levels of transfection in the vHC of SST-Cre mice contribute to the cell-type specificity we observed, PV-Cre transduction in the vHC was comparable to the levels observed in the AMY for both PV-Cre and SST-Cre mice and thus would be expected to influence circuit activity. In agreement with the latter interpretation, a study using the same PV-Cre line and viral strategy found that 8 Hz vHC stimulation was sufficient to influence social discrimination.²⁸

This new framework, wherein the phase relationships of inter-neuron populations across brain structures at specific frequencies define network states that causally influence behavior, raises many important questions for future studies. First, how do these precise phase relationships arise? Second, how does the presence of β -synchrony between GABAergic neuron populations in interconnected brain structures affect information processing to impact approach-avoidance behaviors? The vHC contains multiple non-overlapping output streams that connect it with other nodes of the anxiety network, allowing it to bidirectionally influence anxiety states based on the recruitment of particular vHC subnetworks.^{10–12} The BLA is also hypothesized to transmit anxiety-related information to vHC circuits.^{16–18} Thus, if β -frequency activity in SST+ interneurons entrains BLA output, then synchronization between SST+ inhibitory microcircuits in the BLA and vHC inputs may impact how the vHC integrates BLA inputs, regulating the gain of these anxiety-related inputs and/or their propagation into specific classes of vHC projection neurons. Long-range projections from BLA-SST+ neurons may also rhythmically gate activity within specific vHC microcircuits.

Hippocampus-amygdala β -communication is a cross-species biomarker of anxiety

Previous studies attempting to identify neural signatures of anxiety in rodents have done so by comparing brain activity when mice are in areas that are assumed to be safe with brain activity when mice are in areas assumed to be anxiogenic or examining genetic models with pathological anxiety.^{23,29–31} By contrast, we used neural recordings from wild-type mice in a familiar, home-cage environment to predict their current anxiety level, which was then assayed by subsequent exposure to the EZM. Furthermore, we focused on a biomarker that tracks day-to-day variation in behavior within a single subject (“state”), rather than markers of between-subject differences (“trait”). Although β -frequency oscillations are traditionally associated with sensorimotor processing, increasing evidence has implicated them in a variety of cognitive functions, including decision making, working memory, and reward processing.^{32–35} Recently, β -coherence within the anxiety network, including vHC-BLA interactions, was found to be increased in a mouse model with elevated anxiety,²⁹ suggesting that β -synchrony may contribute to pathological as well as adaptive manifestations of anxiety. An interesting future direction will be investigating how anxiolytic drugs or perturbations that exacerbate anxiety affect this activity pattern.

A simple view is that the EPM comprises “safe” closed arms, “anxiogenic” open arms, and an intermediate center zone. One outstanding ambiguity is whether open-arm exploration reflects periods when the mouse is less anxious (willing to explore) or more anxious (exposed to the open arms). In this context, it is notable that we found that the β -synchrony of SST+ neurons across the BLA and vHC was actually higher in the center zone than in both the closed and open arms and specifically increases just before decisions to avoid the open arms. This suggests that vHC-BLA-SST+ neuron β -synchrony reflects the specific approach-avoidance conflict that occurs in the center zone, rather than more nebulous and anthropomorphic notions of “anxiety” traditionally associated with the open arms.

Modulating synchrony bidirectionally influences anxiety-related behaviors

Although identifying a correlative biomarker can be useful for diagnosis, treatment stratification, or emotional readout, it does not address whether the biomarker is a causal driver of associated behaviors. Here, using optogenetics, we show that β -synchrony between the BLA and vHC bidirectionally influences anxiety-related behaviors in a cell-type-, phase-, and frequency-specific manner. This indicates that approach behaviors in the EPM can be decomposed into at least two stages: “risk assessment” (open-arm probing) and “exploration” (open-arm entry). In this formulation, out-of-phase stimulation is sufficient to promote the first, but not the second, of these stages. This aligns with our observation that vHC-BLA-SST+ neuron β -synchrony is higher in the center zone than in both the open and closed arms. Thus, this pattern of synchronization reflects and causally influences risk-assessment behavior rather than the drive to explore the open arms.

Conclusions

β -synchrony between the AMY and HPC is a cross-species biomarker of emotional state on both long and short timescales. This biomarker is driven by the synchrony of SST+ interneuron populations across these structures. Altering interregional phase relationships between these neurons is sufficient to bidirectionally influence avoidance and risk-assessment behaviors. These results establish that phase relationships between specific neuronal populations in different brain structures are a key determinant of network states encoding emotion. This both informs our understanding of how the brain encodes emotion and may lead to new interventions that modify network states defined by synchrony between specific cell types in order to treat pathological forms of anxiety.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.neuron.2023.12.017>.

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AUTHOR CONTRIBUTIONS

A.D.J. and V.S.S. designed the rodent experiments and analysis. A.D.J. performed all rodent experiments and analyzed the data. J.L.C. and V.S.S. designed the human data analysis, and J.L.C. analyzed the human iEEG data. A.J.P. contributed code for TEMPO signal cleaning. E.F.C. performed the implantations and collected the human iEEG data as part of a previous project directed by E.F.C. and H.E.D. A.D.J. and V.S.S. wrote the manuscript.

DECLARATION OF INTERESTS

V.S.S. is a paid consultant for MapLight Therapeutics.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
AAV5-EF1a-double floxed-hChr2(H134R)-EYFP	Depositing lab: Karl Deisseroth, Stanford	Addgene; 20298-AAV5
AAV5-EF1a-DIO-eNpHR 3.0-EYFP	https://doi.org/10.1016/j.cell.2010.02.037	Addgene; #26966-AAV5
AAV5-EF1a-DIO-eYFP	Depositing lab: Karl Deisseroth, Stanford	Addgene; 27056-AAV5
AAV1-CAG-DIO-Ace2N-4AA-mNeon	Virovek (Plasmid provided by Mark J Schnitzer – Stanford)	N/A
AAV8-nEF-Con/Foff 2.0-ChR2(ET/TC)-EYFP	https://doi.org/10.1016/j.neuron.2020.06.003	Addgene; #137140-AAV8
AAV2-Syn-tdTomato	SignaGen Laboratories	SL100896
AAV1-EF1a-Flpo	https://doi.org/10.1038/nmeth.2996	Addgene; #55637-AAV1
Chemicals, peptides, and recombinant proteins		
Gabazine	Tocris	#1262
CNQX	Tocris	#0190
AP5	Tocris	#0106
Experimental models: Organisms/strains		
SST-Cre mice	The Jackson Laboratory	IMSR_JAX:013044
PV-Cre mice	The Jackson Laboratory	IMSR_JAX:008069
C57BL/6J mice	The Jackson Laboratory	IMSR_JAX:000664
Software and algorithms		
MATLAB	Mathworks	SCR_001622
GraphPad Prism	GraphPad	SCR_002798
ImageJ	ImageJ	SCR_003070
pCLAMP 10	Molecular Devices	SCR_011323
Open Ephys GUI	Open Ephys	SCR_021624
Chronux toolbox	Chronux	SCR_005547
Bonsai	Bonsai	SCR_021512
ANY-maze	Stoelting Co.	SCR_014289
Analysis code	This paper	https://doi.org/10.5281/zenodo.10407015

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Vikaas S. Sohal (vikaas.sohal@ucsf.edu).

Materials availability

This study did not generate any new reagents.

Data and code availability

- All data reported in this paper will be shared by the [lead contact](#) upon reasonable request.
- All original code has been deposited in a Github repository (https://github.com/adj4/Jacksonetal_2024) and is publicly available as of the date of publication. The DOI is listed in the [key resources table](#).
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

All animal care procedures and experiments were conducted in accordance with the National Institutes of Health guidelines and approved by the Administrative Panels on Laboratory Animal Care at the University of California, San Francisco. Mice were housed in a temperature-controlled environment (22–24 °C) with ad libitum access to food and water. Mice were reared in normal lighting conditions (12-h light/dark cycle). Male and female mice aged 10–16 weeks of the following lines were used: Wild-type C57Bl/6 mice, *Sst* < *tm2.1(cre)Zjh* > /J (line 013044; <https://www.jax.org>), and B6;129P2-Pvalb < *tm1(cre)Arbr* > /J (line 008069; <https://www.jax.org>).

METHOD DETAILS

General Surgery

Mice were anaesthetized using isoflurane (2.5% induction, 1–1.5% maintenance in 95% oxygen, flow rate 1L/min) and secured using ear bars in a stereotaxic frame (David Kopf instruments). Body temperature was maintained using a heating pad. The scalp and the periosteum were removed from the dorsal surface of the skull which was then aligned using bregma and lambda as vertical references. For virus surgeries, viruses were injected at a rate of 100nl/min with a 35-gauge microinjection syringe (World Precision Instruments) connected to a microsyringe pump (UMP3 UltraMicroPump, World Precision Instruments). After viral infusion, the needle was kept at the injection site for a minimum of 5mins and then slowly withdrawn. After surgery, animals were allowed to recover on a heated pad until ambulatory.

For LFP only surgeries, 75 μ m tungsten wire electrodes (California Fine Wire) were inserted into the BLA (-1.5 AP, -3.3ML, -4.7 DV) and the vHC (-3.2 AP, -3.2 ML, -4.1 DV). A reference and ground screw was implanted over the cerebellum. Electrodes were initially fixed in place with light activated cement (Ivoclar Vivadent) and then affixed to the skull with Metabond (Parkell). Electrodes and the ground wire were then connected to an electrode interface board (EIB-16, Neuralynx) which was then fixed in place using dental cement.

For TEMPO surgeries, a 2:1 mixture of AAV1-CAG-DIO-Ace2N-4AA-mNeon (Virovek) and AAV2-Synapsin-tdTomato (1.23×10^{12} vg ml⁻¹; SignaGen Laboratories) were injected at the following coordinates: for BLA, -1.5 AP, -3.3 ML, -4.8 and -4.6 DV (150 μ l per site). For vHC, -3.8 AP, -3.2 ML, -4.1 and -4 DV (250 μ l per site) at an 11-degree angle. After virus injection, a custom made optrode was implanted in both the BLA and vHC. Each optrode consisted of a 400/430- μ m (core/outer) diameter, NA = 0.48, fiber implant (Doric Lenses) with a tungsten LFP wire affixed such that the tip of the electrode protruded 200–300 μ m beyond the end of the optic fiber. Behavior was performed at least 5 weeks after injection to allow sufficient time for virus expression.

For channelrhodopsin surgeries, mice were bilaterally injected in both the BLA (-1.5 AP, -3.3 ML, -4.8 and -4.6 DV; 150 μ l per site) and the vHC (-3.8 AP, -3.2 ML, -4.1 and -4 DV; 250 μ l per site at an 11-degree angle) with AAV5-EF1 α -DIO-ChR2-eYFP (Addgene) to selectively target neuronal populations expressing Cre. After injection of virus, a 200/240- μ m (core/outer) diameter, NA = 0.22, was implanted bilaterally in both the BLA and vHC. Behavior was performed at least 4 weeks after injection to allow sufficient time for virus expression.

For halorhodopsin/LFP surgeries, mice were unilaterally injected in both the BLA (-1.5 AP, -3.3 ML, -4.8 and -4.6 DV; 150 μ l per site) and the vHC (-3.8 AP, -3.2 ML, -4.1 and -4 DV; 250 μ l per site at an 11-degree angle) with AAV5-EF1 α -DIO-eNpHR-eYFP (Addgene) to selectively target neuronal populations expressing Cre. After injection of virus, a custom made optrode was implanted in both the BLA and vHC. Each optrode consisted of a 200/240- μ m (core/outer) diameter, NA = 0.22, with a tungsten LFP wire affixed such that the tip of the electrode protruded 200–300 μ m beyond the end of the optic fiber. Behavior was performed at least 4 weeks after injection to allow sufficient time for virus expression.

For halorhodopsin surgeries, mice were bilaterally injected in both the BLA (-1.5 AP, -3.3 ML, -4.8 and -4.6 DV; 150 μ l per site) and the vHC (-3.8 AP, -3.2 ML, -4.1 and -4 DV; 250 μ l per site at an 11-degree angle) with AAV5-EF1 α -DIO-eNpHR-eYFP (Addgene) to selectively target neuronal populations expressing Cre. After injection of virus, a 200/240- μ m (core/outer) diameter, NA = 0.22, was implanted bilaterally in both the BLA and vHC. Behavior was performed at least 4 weeks after injection to allow sufficient time for virus expression.

Behavioral assays

General

Mice were group housed in reversed 12-hr light/dark cycles and all experiments were performed during the dark portion of the cycle. After sufficient time for surgical recovery and/or viral expression, mice underwent multiple rounds of habituation. Each day, mice were brought to the behavioral testing room and transferred to individual cages. These cages were used throughout habituation and testing and acted as individual 'home' cages. Mice were first habituated to the behavioral testing room for 30 mins prior to beginning of any further handling each day. Mice were habituated to touch with 2–3 days of handling for 5 mins each day followed by 2 days of habituation to any required tethers for 10–15 mins.

Elevated zero maze

For LFP only experiments, mice were first recorded in their homecage for 30 mins. Immediately following this, mice were transferred to the elevated zero maze (EZM) for 5 mins. Mice were placed in the entrance of a closed arm, facing into the closed arm. Animals were tested 3 times with a 48hr inter-trial interval.

Elevated plus maze

For LFP only and TEMPO experiments, mice were assessed using the elevated plus maze (EPM) for 15 mins. Animals were placed in the center of the EPM facing a closed arm. For channelrhodopsin experiments, EPM sessions lasted 9 mins. In-phase or out-of-phase channelrhodopsin stimulation (473nm, 3-5mW peak power) were delivered during the second and third three minute epochs and were counterbalanced across animals. For in-phase stimulation, a function generator (Agilent 33500B Series Waveform Generator) connected to two lasers generates simultaneous 20Hz trains of 5ms pulses which are coupled to fiber optic implants through a 200 μ m diameter dual fiber optic patchcord with a guiding socket (Doric Lenses). For out-of-phase stimulation, each 473nm laser was connected to a separate function generator. The triggering of these function generators was offset by 25ms, producing a phase difference of 180°. Real-time light delivery was based on location within the apparatus, with the stimulation zone restricted to the center zone and the proximal third of the closed and open arms. For halorhodopsin experiments, 532nm stimulation (4-6mW peak power) was delivered in based on location within the apparatus, with the stimulation zone restricted to the center zone and the proximal third of the closed and open arms.

Cross maze

Experiments were performed with the same experimental design as described for EPM with optogenetic stimulation but with a cross shaped maze consisting of 4 closed arms. Animals were placed in the center of the cross maze facing a closed arm on the axis that was classified as 'Closed 1'. 'Closed 2' arms were classified as the two arms that run perpendicular that axis, which are the geographical equivalent of the open arms in the EPM.

Real-time place preference (RTPP)

RTPP was performed with an apparatus containing two identical sides with opaque white walls. On day 1, mice were habituated to the apparatus for 15mins. On day 2, mice were allowed to freely explore the two chambers and the amount of time spent in each chamber was quantified. On day 3, 532nm stimulation (4-6mW peak power) was delivered in one side of the chamber only in a location-dependent manner.

Analysis

For location-based optogenetic stimulation experiments, animal location was tracked using any-maze software. For all other behavioral testing, Bonsai software and custom MATLAB scripts were used to measure animal locomotion speed and location.

LFP recordings

Recordings and analysis

LFP data were sampled at 2KHz and band-pass filtered from 0.1-200Hz using an Open Ephys acquisition board (Open Ephys) connected to an Intan RHD 16-channel headstage (Intan). Electrode placements were verified histologically. Analysis of LFP data was performed using custom MATLAB code. Data was assessed in the following frequency bands: theta band (4-12Hz), beta band (13-30Hz), low gamma band (30-50Hz) and high-gamma band (65-90Hz). Coherence was calculated in 10s non-overlapping windows using MATLAB's *mscohere* function. Variance of coherence was calculated with a sliding 60s window. For β -coherence burst analysis, multi-taper time-frequency coherence measures were generated with *cohgramc* function from the Chronux toolbox (<http://chronux.org>). β -coherence bursts were classified as time and frequency-continuous periods of coherence greater than 80% maximum coherence. For EPM runs, β -coherence burst probability was assessed in 500ms windows from 5s before to 5s after the animal exited the center zone. Data was z-scored to probability across the full recording session.

TEMPO recordings

Optical apparatus

A fiber-optic stub (400- μ m core, NA = 0.48, low-autofluorescence fiber; Doric Lenses, MFC_400/430-0.48_2.8mm_ZF1.25_FLT) was stereotactically implanted in each targeted brain region. A matching fiber-optic patch cord (Doric Lenses, MFP_400/430/1100-0.48_2m_FC-ZF1.25) provided a light path between the animal and a miniature, permanently aligned optical bench, or 'mini-cube' (Doric Lenses, FMC5_E1(460-490)_F1(500-540)_E2(555-570)_F2(580-680)_S). A single fiber was used to both deliver excitation light to and collect emitted fluorescence from each recording site. The far end of the patch cord and each 1.25-mm-diameter zirconia optical implant ferrule were cleaned with isopropanol before each recording and then securely attached via a zirconia sleeve.

The mini-cube optics allow for the simultaneous monitoring of two spectrally separated fluorophores, with dichroic mirrors and cleanup filters chosen to match the excitation and emission spectra of the voltage sensor and reference fluorophores in use ('mNeon' voltage sensor channel: excitation, 460–490 nm, emission, 500–540 nm; 'Red' control fluorophore: excitation, 555–570 nm, emission, 580–680 nm). The mini-cube optics were sealed and permanently aligned, and all five ports (sample to animal, two excitation lines and two emission lines) were provided with matched coupling optics and FC connectors to allow for a modular system design.

Excitation light for each of the two color channels was provided by a fiber-coupled LED (center wavelengths, 490 nm and 565 nm, Thorlabs, M490F3 and M565F3) connected to the mini-cube by a patch cord (200 μ m, NA = 0.39; Thorlabs, M75L01). Using a smaller diameter for this patch cord than for the patch cord from the cube to the animal is critical to reduce the excitation spot size on the

output fiber face and thus avoid cladding autofluorescence. LEDs were controlled by a four-channel, 10-kHz-bandwidth current source (Thorlabs, DC4104). LED current was adjusted to give a final light power at the animal (averaged during modulation, see below) of approximately 200 μ W for the mNeon channel (460–490-nm excitation) and 100 μ W for the Red channel (555–570-nm excitation).

Each of the two emission ports on the mini-cube was connected to an adjustable-gain photoreceiver (Femto, OE-200-Si-FC; bandwidth set to 7 kHz, AC coupled, 'Low' gain of $\sim 5 \times 10^7$ V/W) using a large-core high-NA fiber to maximize throughput (600- μ m core, NA = 0.48 (Doric lenses, MFP_600/630/LWMJ-0.48_0.5m_FC-FC)).

Note that, for dual-site recordings, two completely independent optical setups were employed, with separate implants, patch cords, mini-cubes, LEDs, photoreceivers and lock-in amplifiers.

Modulation and lock-in detection

At each recording site, each of the two LEDs was sinusoidally modulated at a distinct carrier frequency to reduce crosstalk due to overlap in fluorophore spectra. The corresponding photoreceiver outputs were then demodulated using lock-in amplification techniques. A single instrument (Stanford Research Systems, SR860) was used to generate the modulation waveform for each LED and to demodulate the photoreceiver output at the carrier frequency. To further reduce crosstalk between recording sites, distinct carrier frequencies were used across at each site (mNeon site 1: 2 kHz; mNeon site 2: 2.5 kHz; Red site 1: 3.5 kHz; Red site 2: 4 kHz). Low-pass filters on the lock-in amplifiers were selected to reject noise above the frequencies under study (cascade of four Gaussian FIR filters with 84-Hz equivalent noise bandwidth; final attenuation of signals were approximately -1 dB (89% of original magnitude) at 20 Hz, -3 dB (71% of original magnitude) at 40 Hz and -6 dB (50% of original magnitude) at 60 Hz).

Recording

Analog signals were digitized by a multichannel real-time signal processor (Tucker-Davis Technologies, RX-8). The commercial software Synapse (Tucker-Davis Technologies) running on a PC was used to control the signal processor and write data streams to disk. Lock-in amplifier outputs were digitized at 3 kHz. Signals were downsampled to 2KHz for further analysis.

Analysis

Analysis of TEMPO signals was performed using the signal processing toolbox and custom written MATLAB code. All four signals from the from recording sessions (vHC mNeon, vHC tdTomato, BLA mNeon, BLA tdTomato) were filtered around the frequency of interest using the Matlab function *filfilt*. mNeon signals for each region were corrected by subtraction of the robust linear regression model (function *robustfit*) of the tdTomato and mNeon signals across 1s windows. For correlation across full sessions or in zones of the EPM, zero-phase lag correlation was calculated over 250ms non-overlapping windows. For time-locked analysis during EPM zone transitions, 80% overlapping 1s windows were analyzed from 5s before to 5s after the animal exited the center zone. For assessment of TEMPO signal synchrony during β -coherence bursts, LFP signals were first analyzed as described to identify the onset of β -coherence bursts. Analysis was restricted to β -bursts with a duration of at least 500ms and where the correlation of β -filtered vHC and BLA LFP signals exceeded 0.5. Cross-correlation of signals during the first 500ms of each β -coherence burst was performed and averaged across bursts for each animal. The phase locking value was then calculated across animals for each group using the peak correlation and phase offset for each mouse. This value was compared to 1000 PLVs generated from randomly selected timepoints when β -coherence bursts were not occurring. Time-locked TEMPO signal analysis was then performed using a correlation window of 500ms with 90% overlap. Data correlations were compared to 100 correlations from randomly shuffled data.

Histology and imaging

All mice used for behavioral and imaging experiments were anesthetized with Euthazol and transcardially perfused with ice-cold 0.01 M PBS followed by ice-cold 4% paraformaldehyde (PFA) in PBS. Brains were extracted and stored in 4% PFA for 24 h at 4 °C before being stored in PBS. Next, 75 μ m slices were obtained on a Leica VT100S and mounted on slides using aqueous mounting medium containing DAPI (Vectashield Plus Antifade mounting medium, Vector labs). Sections were imaged using a BZ-X fluorescence microscope (Keyence). Cell counts were performed with 3 randomly selected SST-Cre and PV-Cre mice. For each mouse, 3 slices from each brain region were imaged and cell counts performed within 3 randomly placed 200 μ m² areas. Averages for each slice were calculated. Cell counts were automated using Fiji software (<https://imagej.net/software/fiji/>).

Human recordings

Intracranial electroencephalography (iEEG) recordings were obtained from 13 human subjects (n=9 Male, n=4 Female) with treatment-resistant epilepsy with surgically implanted semi-chronic intracranial electrodes (for sites included in this study: Ad-Tech 4-contact strip or depth, 10mm center-to-center spacing, 2.3mm exposed diameter; Ad-Tech 10-contact depth, 6 or 5mm center-to-center spacing) for the clinical purpose of seizure localization. All procedures were approved by the University of California, San Francisco Institutional Review Board. All subjects gave written informed consent to participate in the study before surgery. 6 subjects had left hemisphere implantations, 7 had right hemisphere implantations). Subjects' psychological state was evaluated up to four times daily during their hospitalization period prompted by a research assistant, using a tablet-based, custom-designed questionnaire called the Immediate Mood Scaler (Nahum et al.³⁶) (IMS). Details of psychological testing, electrophysiologic recording, data processing, and coherence calculations were reported previously⁴.

AMY-HPC beta (β , 13–30 Hz) coherence was calculated in 5 second bins with a 0.5 second sliding window over a 20 minute period centered on an IMS score. Coherence variance was calculated as the mean of the rolling variance (bin = 10) over the mood score window. β -coherence bursts were defined as beginning when coherence increases by a set magnitude (50% of the max coherence during the 20 minute IMS window) above the mean coherence of a 30 second trailing window (e.g. mean coherence during the

previous 30 seconds). β -coherence bursts end when coherence drops below 50% of the max coherence. β -coherence bursts were quantified for each time window by summing the area under the curve (AUC) for all bursts, counting the number of bursts, and finding the mean burst duration.

Statistics were performed using the SciPy stats package. Linear regressions between IMS and coherence variance, coherence mean, and sum AUC were performed. As was a linear regression for sum AUC and coherence variance. Student's t-tests were performed to compare the sum AUC, number of bursts, and burst duration between the top and bottom quartile of IMS scores.

Intracellular recordings

Viral injections

SST-Cre/Ai14 mice were bilaterally injected in the BLA (-1.5 AP, -3.3 ML, -4.8 and -4.6 DV; 150 μ l per site) with AAV8-nEF-Con/Foff 2.0-ChR2(ET/TC)-EYFP (Addgene). Additionally, the vHC (-3.8 AP, -3.2 ML, -4.1 and -4 DV; 250 μ l per site at an 11-degree angle) was injected with AAV1-EF1a-Flpo (Addgene) prevent any potential trans-synaptic transfer of channelrhodopsin to target SST+ cells in the vHC. Recording was performed at least 4 weeks after injection to allow sufficient time for virus expression.

Tissue slicing

Mice were anaesthetized using isoflurane and their brains quickly dissected out and placed in ice-cold (<4°C) slicing aCSF solution (in mM: NaCl 86, NaH₂PO₄ 1.2, KCl 2.5, NaHCO₃ 25, D-glucose 25, Sucrose 75, CaCl₂ 0.5, MgCl₂ 7) saturated with continuous carbogen bubbling (95% O₂, 5% CO₂). 250 μ m coronal slices containing vHC were cut using a vibratome (VT1200S, Leica) and transferred to a holding chamber containing recording aCSF solution (in mM: NaCl 124, NaH₂PO₄ 1.2, KCl 2.5, NaHCO₃ 25, D-glucose 20, CaCl₂ 2, MgCl₂ 1) where they were maintained at 36°C for 30mins. Slices were left to recover at room temperature for a further 30mins prior to the start of recording.

Recording

Slices were transferred to a recording chamber of an upright microscope (BX51WI, Olympus). They were perfused with carbogenated recording aCSF solution at a rate of 2-3ml/min heated to 31°C. Neurons were visualized using DIC optics and SST+ cells in the vHC were identified by tdTomato fluorescence using a 40x water-immersion objective. Patch electrodes (2-4M Ω) were pulled from borosilicate capillary glass with a 1mm external diameter (P-2000, Sutter Instruments). Electrodes were back-filled with an internal solution containing: (in mM) K-Gluconate 120, KCl 20, HEPES 10, NaCl 4, Mg-ATP 4, Na-GTP 0.3, pCreatine 10). Data were recorded using a Multiclamp 700B amplifier (Molecular Devices). Pipette capacitance was compensated before achieving whole-cell configuration. Series resistance was monitored throughout the recording and experiments were terminated if series resistance exceeded 25M Ω .

To measure optogenetically evoked potentials in vHC-SST+ neurons, ChR2 was stimulated using 5ms long 480nm light pulses (max power 3mW) using a Lambda FLED system (Sutter Instruments). Responses were measured at 0mV in recording aCSF solution and combined with the following compounds: 6-cyano-7-nitroquinoxaline-2,3-dione disodium salt hydrate (CNQX), d-2-amino-5-phosphonopentanoic acid (APV) and 2-(3-carboxypropyl)-3-amino-6-(4-methoxyphenyl)-pyridazinium bromide (Gabazine; GBZ) (Tocris).

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistics were calculated using MATLAB code or Graphpad Prism. All statistical parameters, including test statistics, corrections for multiple comparisons and sampling are stated in the main text. Data was tested for normality with the Shapiro-Wilk test. All tests are two-sided and error bars represent standard error of the mean. Sample sizes were based on prior studies. All tests and n numbers are summarized in [Table S1](#).